Isheja Ladouce

Part 3 : Gradient Descent

Updating m and b (iteration 1)

Given

$n = 2$, α (learning rate) $= 0.01$, $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$, $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $b = 1$

* the Model : $\hat{y} = \underbrace{m_1 x_1 + m_2 x_2}_{\text{Model weights}} + b$ ↳ bias

(x_1, x_2: Input features)

* Mean Squared Error (MSE)

$$J = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

Gradients :

Weight gradient

$$\frac{\partial J}{\partial m_j} = \frac{2}{n} (\hat{y}_i - y_i) x_{ij}$$

Bias gradient

$$\frac{\partial J}{\partial b} = \frac{2}{n} (\hat{y}_i - y_i)$$

Update formulas :

$$m_{new} = m - \alpha \nabla m$$

$$b_{new} = b - \alpha \nabla b$$

} formulas

* Iteration 1: Initial prediction

$$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad b = 1 \qquad X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$\hat{y} = mx_1 + mx_2 x_2 + b$$

for (1,3): $\hat{y}_1 = (-1)(1) + 2(3) + 1 = -1 + 6 + 1 = 6$

for (4,10): $\hat{y}_2 = (-1)(4) + 2(10) + 1 = -4 + 20 + 1 = 17$

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

* Error $(e = \hat{y} - y)$

$$e = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

* Calculating gradients, multiplying by $2/n$ $(n = 2) = 1$

- $\nabla m = \frac{2}{n} X^T e$ $\quad \frac{2}{n} = 1, \; X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, \; X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$

$$X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} \cdot 1 = \begin{bmatrix} 1(1) + 4(11) \\ 1(3) + 10(11) \end{bmatrix} = \begin{bmatrix} 1 + 44 \\ 3 + 110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\nabla m = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

- Bias gradients

$$\nabla b = \frac{2}{n} (\hat{y}_i - y_i)$$

$$= 1 \frac{2}{2} (1 + 11) = 1 + 11$$

$$\nabla b = 12$$

* Update the parameters

Update m (weights) $\quad m_{new} = m - \alpha \nabla m$

$$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$= \begin{bmatrix} -1 - 0.01(45) \\ 2 - 0.01(113) \end{bmatrix}$$

$$m_{new} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

- Update the bias

$$b_{new} = b - \alpha \nabla_b$$

$$= 1 - 0.01(12) = 1 - 0.12$$

$$b_{new} = 0.88$$

Iteration 1 $m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$, $b = 0.88$

Scond iteration

$$m_1 = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$b_1 = 0.88$, $n = 2$

$$\hat{y} = m_1 x + b_1 (1)$$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 1(-1.45) + 3(0.87) \\ 4(-1.45) + 10(0.87) \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} -1.45 + 2.61 \\ -5.8 + 8.7 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 1.16 \\ 2.9 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$$

$$e = \hat{y} - y$$

$$e = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e = \begin{bmatrix} 2.04 - 5 \\ 3.78 - 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$\frac{\partial MSE}{m} = \frac{2}{n} X^T e$$

$$X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-2.96) + 4(-2.22) \\ 3(-2.96) + 10(-2.22) \end{bmatrix}$$

$$= \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

$$\frac{\partial MSE}{m} = \frac{2}{2} \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

$$\frac{\partial MSE}{b} = -\frac{2^1}{2_1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} (-2.96 + (-2.22))$$

$$\frac{\partial MSE}{L} = -5.18$$

Scanned with CamScanner

$$m_2 = m_1 - \alpha \frac{\partial MSE}{m}$$

$$m_2 = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

$$m_2 = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - \begin{bmatrix} -0.1184 \\ -0.3108 \end{bmatrix}$$

$$m_2 = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$$

$$b_2 = b_1 - \alpha \frac{\partial MSE}{b}$$

$$b_2 = 0.88 - 0.01 (-5.18)$$

$$b_2 = 0.9318$$

Scanned with CamScanner

Iteration 3 - UNIDEAL BONEHEAD Regina

$m_{new} = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$ $b = \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$

1. Predict

$\hat{y} = mx + b$

$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} = \begin{bmatrix} -1.3316 + 3.5424 \\ -5.3264 + 11.808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$

2. Errors

$e = y - \hat{y} = \begin{bmatrix} 5 \\ 8 \end{bmatrix} - \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} = \begin{bmatrix} 1.8574 \\ 0.9121 \end{bmatrix}$

$MSE = \frac{\sum (y - \hat{y})^2}{n}$

$(y-\hat{y})^2 = e^T \cdot e = [1.8574 \ 0.9121] \begin{bmatrix} 1.8574 \\ 0.9121 \end{bmatrix} =$

$= 3.449734476 + 1.997769956$

$MSE = 2.72381716 \simeq 2.7238$

3. Gradients

$\frac{\partial MSE}{\partial m} = \left[\frac{1}{n}(y - mx - b)^2\right]'$

$= \left(\frac{1}{n}\right)'(y - mx - b)^2 + \frac{1}{n}\left[(y - mx - b)^2\right]'$

$$\frac{\partial MSE}{\partial m} = \frac{1}{n}\left((y-mx-b)^2\right)'$$

let $y - mx - b = u$

$$(u^2)' = 2uu'$$

$$u' = (y-mx-b)'$$

$$= 0 - x - 0$$

$$u' = -x$$

then $(u^2)' = 2x\,(y-mx-b) \times (-x)$

$$\frac{\partial MSE}{\partial m} = -\frac{2x}{n}\,(y-mx-b)$$

$$\frac{\partial MSE}{\partial b} = \left(\frac{1}{n}\,(y-mx-b)^2\right)'$$

$$= \left(\frac{1}{n}\right)'(y-mx-b)^2 + \left(\frac{1}{n}\right)\left[(y-mx-b)^2\right]'$$

let $u = y - mx - b$

$$(u^2)' = 2uu'$$

$$u' = \frac{dy}{db} = 0 - 0 - 1$$

$$\frac{\partial MSE}{\partial b} = 2 \cdot (y-mx-b) \cdot (-1) \cdot \frac{1}{n}$$

$$\frac{\partial MSE}{\partial b} = -\frac{2}{n} \cdot 1 \cdot (y-mx-b)$$

Find the values:

$$\frac{\partial MSE}{\partial m} = -\frac{2}{2} \cdot \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1.8574 \\ -1.4134 \end{bmatrix}$$

CamScanner

$$\frac{\partial MSE}{\partial m} = -1 \begin{bmatrix} -3.7862 \\ -8.5618 \end{bmatrix} = \begin{bmatrix} 3.7862 \\ 8.5618 \end{bmatrix}$$

$$\frac{\partial MSE}{\partial b} = -\frac{2}{2} \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1.8574 \\ -1.4134 \end{bmatrix}$$

$$= -1 \begin{bmatrix} 1.8574 \\ -1.4134 \end{bmatrix}$$

$$\frac{\partial MSE}{\partial b} = -1 \begin{bmatrix} 1.8574 - 1.4134 \end{bmatrix}$$

$$= -0.444 \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \frac{\partial MSE}{\partial b} = \begin{bmatrix} 0.444 \\ 0.444 \end{bmatrix}$$

4. Update

$$m_{new} = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} - 0.01 \begin{bmatrix} 3.7862 \\ 8.5618 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} - \begin{bmatrix} 0.037862 \\ 0.085618 \end{bmatrix}$$

$$m_{new} = \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix}$$

$$b_{new} = b_{old} - \alpha \frac{\partial MSE}{\partial b}$$

$$= \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} - 0.01 \begin{bmatrix} -0.444 \\ 0.444 \end{bmatrix}$$

$$b_{new} = \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix}$$

CamScanner